

Immature granulocyte: an early predictor of strangulated adhesive small bowel obstruction.

Su Q, Wang Q, Cao Y

BMC gastroenterology (2025) • PMID 41152714 • PMC12570669 • DOI 10.62713/aic.3794

Objective

Early identification of strangulation in adhesive small bowel obstruction (SBO) is crucial for emergency triage and immediate surgical intervention. This study aimed to explore the predictive value of the immature granulocyte (IG) count in assessing strangulation in adhesive SBO.

Methods

This was a retrospective analysis of patients with adhesive SBO, diagnosed surgically or by abdominal computed tomography (CT) from May 2022 to April 2025, comparing strangulated (SG) and non-strangulated (NSG) cases. Baseline levels of the IG count and related inflammatory biomarkers were analyzed using logistic regression analysis and receiver operator characteristic (ROC) curves to assess the predictive value for strangulation.

Results

250 eligible patients in the SG (n = 167) and NSG (n = 83) groups were enrolled in this study. Compared to the NSG group, the SG group demonstrated significantly higher levels of white blood cell count (WBC), neutrophil (NEU), neutrophil-to-lymphocyte ratio (NLR), and IG count with statistical differences (all $P < 0.001$). In the analysis of logistic regression and ROC, the IG count demonstrated an independent risk for predicting strangulated adhesive SBO (OR 15.41, 95%CI 8.07–31.59, $P < 0.001$). The IG count had the highest area under the ROC curve (AUC) of 0.946 among the enrolled inflammatory biomarkers, particularly within 6 h from onset to hospital (AUC 0.961, 95% CI, 0.90–1.00). It showed an optimal cutoff value of $0.280 \times 10^9/L$ with a sensitivity of 89.50% and specificity of 100.00% for identifying strangulation in adhesive SBO.

Conclusion

IG count was an early predictor for quick detection of strangulation in adhesive SBO within 6 h from onset to hospital. This finding may fill the clinical gap in recognizing intestinal ischemia at an early stage prior to imaging examination.

Keywords

Introduction

Strangulated adhesive small bowel obstruction (SBO) is a critical surgical emergency in the acute abdominal spectrum. Among all adhesive SBO, 5–10% may progress to strangulation [1]. Delayed recognition and surgical intervention carry significant risks of catastrophic sequelae, including bowel necrosis (30–60%), perforation (15–25%), sepsis (25–40%), and multiorgan failure (15–30%) [2, 3]. It exhibits a 5 to 10-fold increase in mortality compared with simple SBO [4]. Early and prompt identification of strangulation is particularly important. Advanced imaging modalities and exploratory laparotomy still occupy the main positions for definitive confirmation in clinical practice. Computed tomography (CT) is now established as the best imaging technique for the initial assessment of patients suspected of having strangulation in adhesive SBO [4]. However, the positive predictive value of spectral CT for intestinal ischemia is only 50% at an early stage [5]. Although intraoperative findings and histopathological confirmation serve as the diagnostic gold standard for strangulated adhesive SBO, their clinical utility is inherently limited by the invasive nature of surgical exploration. Therefore, reliable serum biomarkers may be a novel option for facilitating timely interventions.

Most studies have extensively explored serum biomarkers for predicting strangulation in intestinal obstruction. Traditional inflammatory markers, such as white blood cell count (WBC), neutrophil (NEU), C-reactive protein (CRP), and high-sensitivity CRP (hs-CRP), are typically elevated during advanced intestinal necrosis, whereas their early diagnostic utility decreases under confounding inflammatory conditions [6–8]. In addition, metabolic indicators, such as lactate (>2.5 mmol/L), suggested intestinal ischemia [9]. However, it interferes easily with shock or dehydration and usually increases after 6 h of post-mucosal ischemia [2]. As indicated in a study by Zhou Y et al., D-dimer (≥ 1.5 $\mu\text{g/mL}$) showed high specificity (98%) for identifying intestinal necrosis, but it was also elevated during dehydration [10].

In recent years, laboratory identification of strangulation in SBO has undergone a transition from “non-specific inflammatory markers” to “mechanism oriented biomarkers”. Serum homocysteine (HCY) strongly correlates with irreversible transmural intestinal necrosis with a cut-off value of 15.53 $\mu\text{mol/L}$, a sensitivity of 89.7%, and specificity of 83.0% [11]. However, it had a similar limitation with non-specific traditional inflammatory markers that HCY only started to increase at 6–8 h of post-ischemia, which significantly constrains early detection of strangulation. Surprisingly, immature granulocytes (IG) count, a convenient and easily obtainable routine blood marker, can rise within 2 h of intestinal ischemia, preceding other hematologic markers [12]. As early as 2022 and 2024 years, the groups of Durak et al. and Çaparlar et al. validated the role of IG count in predicting mesenteric ischemia and intestinal volvulus necrosis with a sensitivity of 76.2–74.1% and specificity of 85.7–98.2%, respectively [13, 14]. Hitherto, no study has evaluated the predictive value of IG count for strangulated adhesive SBO.

In this study, we explored the differences in clinical variables and laboratory markers among strangulated adhesive SBO cases and assessed the predictive value in them.

Materials and methods

Study design and population

This study was reviewed and approved by the Ethics Committee of the Third Affiliated Hospital of the Xinxiang Medical University (XXMU). And the requirement of written informed consent from patients was formally waived by Ethical committee of the Third Affiliated Hospital of XXMU, for the reasons that this study performed the retrospective review of de-identified patient data and ensure consistency with institutional approvals. All processes were consistent with relevant ethical guidelines such as the Declaration of Helsinki [15].

This retrospective study enrolled patients with SBO between May 2022 and April 2025. They were from the departments of gastroenterology, emergency surgery, and general surgery. The patients included in this study were 16–80 years of age. They were diagnosed with adhesive SBO based on abdominal CT and were highly suspected to have intestinal ischemia and strangulation according to CT findings of enhancement of the intestinal wall, closed loop sign, vortex sign, intestinal wall, or portal vein gas accumulation [16]. The gold standard was based on intraoperative performance by experienced surgeons during emergency or elective procedures, including intestinal cyanosis, necrosis or darkening, loss of wall elasticity, serosal edema, and absent peristalsis [17]. And they were excluded if they were: (1) not undergoing CT scan; (2) age < 16 or > 80 years; (3) SBO caused by tumor, hernia, inflammatory bowel disease (IBD), etc.; (4) other concurrent infections, such as abdominal, respiratory, and urinary infection; (5) combined with serious hepatorenal insufficiency; and (6) received any medication known to significantly influence leukocyte counts within the 4 weeks prior to enrollment, including antibiotics (e.g., Penicillins, Cephalosporins), corticosteroids (e.g., prednisone, methylprednisolone), Granulocyte colony-stimulating factors (G-CSF), non-steroidal anti-inflammatory drugs (NSAIDs), immunosuppressants (e.g., cyclosporine, methotrexate).

Data collection

Data were extracted from our institutional electronic medical record. All patient data were anonymized and de-identified prior to collection and analysis. The cohort was divided into two groups, stratified by strangulation status: strangulation group (SG), with intraoperative findings of strangulation (e.g.,

ischemic, vascular compromise), and non-strangulation group (NSG), either with intraoperative confirmation of viable bowel (adhesive obstruction without ischemia/necrosis), or who was the absence of signs of ischemia on imaging and successfully managed conservatively, such as nil by mouth, nasogastric decompression, aggressive fluid resuscitation, and empirical intravenous antibiotics, etc. The following data were extracted: age, sex, time of onset to hospitalization, main symptoms, initial vital signs, and treatment management. Preoperative or admission laboratory parameters were also collected, including WBC count, NEU, neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), and IG count. All blood samples in this study were analyzed using the same automated hematology analyzer (Mindray, BC-7500CS) and standardized operating procedure in our institution, ensuring internal consistency of these parameters. The reference interval provided by the manufacturer: WBC count ($3.5\text{--}9.5 \times 10^9/\text{L}$), NEU ($1.8\text{--}6.3 \times 10^9/\text{L}$), lymphocyte count ($1.1\text{--}3.2 \times 10^9/\text{L}$), platelet count ($125\text{--}350 \times 10^9/\text{L}$), and IG count ($\leq 0.029 \times 10^9/\text{L}$).

Statistical analysis

All of data had been quantitatively evaluated, especially missing variables before analysis. When the missing proportion less than 2%, which was considered a completely random missing covariate, a complete case analysis can be performed. Continuous variables were presented as mean $\pm$ standard deviation ($M \pm SD$). Normality was assessed using the Shapiro-Wilk test. Normally distributed continuous variables were compared between the SG and NSG groups using independent sample t-tests; non-normally distributed variables were compared using Mann-Whitney U tests. Categorical variables were expressed as frequencies and percentages (%) and were analyzed using either the chi-squared test or Fisher's exact test, as appropriate. Binary logistic regression analysis was used to identify key predictors of strangulation obstruction, expressed as odds ratios (OR) with 95% confidence intervals (95%CI). Variables with $P < 0.10$ in univariate analyses or clinical relevance were entered into the multivariate model using backward stepwise elimination. Collinearity test of variables were analyzed by SPSS(version 24.0) after univariate and multivariate analyses. Variance inflation factor (VIF) values are significantly less than 5, indicating that there is no serious multicollinearity problem among the independent variables in the model. The diagnostic performance of significant predictors for detecting strangulation was evaluated by using receiver operating characteristic (ROC) curve analysis. The area under the ROC curve (AUC) was used to quantify overall discriminative ability. Optimal cut-off values were determined by maximizing the Youden index ($J = \text{sensitivity} + \text{specificity} - 1$) [18]. The sensitivity (Sen, %) and specificity (Spe, %) have also been reported. Diagnostic utility was further assessed by using the positive likelihood ratio ($LR + = \text{Sen}/(1 - \text{Spe})$) and negative likelihood ratio ($LR - = (1 - \text{Sen})/\text{Spe}$) [19]. Differences were considered statistically significant at $P < 0.05$.

Given the large observed effect size (Cohen's $d = 2.0$, 95% CI 0.52–0.69) for the primary outcome, a post-hoc power analysis about IG count predicting strangulation was conducted using SPSS software (version 24.0). And it confirmed that the achieved statistical power exceeded 99% for the primary comparison at an alpha level of 0.05. Besides that we used K-fold cross validation to address the overfitting of our results ($K = 10$).

Results

Study characteristics

A total of 293 patients with SBO were identified between May 2022 and April 2025. Finally, 250 eligible patients were enrolled in our study according to the inclusion and exclusion criteria, and divided into SG ($n = 167$) and NSG ($n = 83$) groups (Fig. 1). As Table 1 shows, in terms of age, sex, and body mass index (BMI), there were no statistically differences between the SG and NSG groups (all $P > 0.05$). The mean time from symptom onset to hospital was 12.53 h in the SG group, significantly earlier than in the NSG group (22.47 h, $P < 0.001$). Significant differences were observed in initial symptoms, abdominal signs, and treatment strategies between the SG and NSG groups. Vomiting was less prevalent in the SG group than in the NSG group (34.73% vs. 56.63%, $P < 0.001$), while abdominal distension was more common (89.22% vs. 78.31%,

$P < 0.05$). The incidence of abdominal tenderness was significantly higher in the NSG group than in the SG group (77.11% vs. 62.87%, $P < 0.05$). Notably, the treatment strategies differed substantially. Patients in the SG group exclusively underwent emergency operations (100% vs. 2.41% in the NSG group, $P < 0.001$), whereas those in the NSG group primarily received conservative management (84.34% vs. 0.00%, $P < 0.001$). Finally, there was no significant difference among two groups, including abdominal pain (97.60% vs. 98.80%, $P > 0.05$), obstipation (75.45% vs. 79.52%, $P > 0.05$), abdominal enlargement (83.23% vs. 78.31%, $P > 0.05$), and rebound tenderness (12.57% vs. 6.02%, $P > 0.05$). Fig. 1 The flowchart of the enrolled study patients

Table 1 Baseline characteristics and treatment strategies of SG and NSG groups

Variables	SG (n = 167)	NSG (n = 83)	P-value
Age (years, M $\pm$ SD)	45.91 $\pm$ 1.33	47.00 $\pm$ 1.86	0.64
Sex (n, male/female)	97/70	49/34	0.89
BMI (kg/m ² , M $\pm$ SD)	25.75 $\pm$ 0.41	26.23 $\pm$ 0.64	0.51
Time of onset to hospital (hours, M $\pm$ SD)	12.53 $\pm$ 0.56	22.47 $\pm$ 1.15	< 0.001
<6 (n, percentage)	129 (77.25%)	78 (93.98%)	< 0.001
≥ 6 (n, percentage)	38 (22.75%)	5 (6.02%)	< 0.001
Symptoms (n, percentage)			
Abdominal pain	163 (97.60%)	82 (98.80%)	0.53
Vomiting	58 (34.73%)	47 (56.63%)	< 0.001
Abdominal distension	149 (89.22%)	65 (78.31%)	0.02
Obstipation	126 (75.45%)	66 (79.52%)	0.48
Abdominal signs (n, percentage)			
Enlargement	139 (83.23%)	65 (78.31%)	0.35
Tenderness	105 (62.87%)	64 (77.11%)	0.04
Rebound tenderness	21 (12.57%)	5 (6.02%)	0.11
Treatment (n, percentage)			
Conservative	0 (0.00%)	70 (84.34%)	< 0.001
Elective operation	0 (0.00%)	11 (13.25%)	< 0.001
Emergency operation	167 (100%)	2 (2.41%)	< 0.001

SBO small bowel obstruction, SG Strangulated group, NSG No-strangulated group, M $\pm$ SD mean $\pm$ standard deviation, BMI body mass index

The flowchart of the enrolled study patients

Baseline characteristics and treatment strategies of SG and NSG groups

SBO small bowel obstruction, SG Strangulated group, NSG No-strangulated group, M $\pm$ SD mean $\pm$ standard deviation, BMI body mass index

There were statistically significant differences in the serum inflammatory markers between the SG and NSG groups. As Table 2 shown, compared to the NSG group, the SG group demonstrated significantly elevated WBC ($\times 10^9/L$, 13.40 $\pm$ 0.28 vs. 9.38 $\pm$ 0.34, $P < 0.001$), NEU ($\times 10^9/L$, 10.22 $\pm$ 0.24 vs. 7.21 $\pm$ 0.30, $P < 0.001$), NLR (7.17 $\pm$ 0.50 vs. 4.57 $\pm$ 0.48, $P < 0.001$), and IG count ($\times 10^9/L$, 0.76 $\pm$ 0.01 vs. 0.16 $\pm$ 0.03, $P < 0.001$). Furthermore, within less than 6 h from onset time to hospital, the level of IG count in the SG group was also significantly higher than that in the NSG group ($\times 10^9/L$, 0.70 $\pm$ 0.35 vs. 0.16 $\pm$ 0.08, $P < 0.001$). In contrast, PLR showed no statistically significant difference between the two groups (184.64 $\pm$ 14.51 vs. 138.88 $\pm$ 18.13, $P > 0.05$).

Table 2 Laboratory parameters of the patients enrolled at admission or pre-operation

Parameters (M $\pm$ SD)	SG (n = 167)	NSG (n = 83)	P-value
WBC ($\times 10^9/L$)	13.40 $\pm$ 0.28	9.38 $\pm$ 0.34	< 0.001
NEU ($\times 10^9/L$)	10.22 $\pm$ 0.24	7.21 $\pm$ 0.30	< 0.001
NLR	7.17 $\pm$ 0.50	4.57 $\pm$ 0.48	< 0.001
PLR	184.64 $\pm$ 14.51	138.88 $\pm$ 18.13	> 0.05
IG ($\times 10^9/L$) (all time)	0.76 $\pm$ 0.01	0.16 $\pm$ 0.03	< 0.001
IG ($\times 10^9/L$) (<6 h)	0.70 $\pm$ 0.35	0.16 $\pm$ 0.08	< 0.001

SG Strangulated group, NSG No-strangulated group, M $\pm$ SD mean $\pm$ standard deviation, WBC white blood cell, NEU neutrophils, NLR neutrophil-to-lymphocyte ratio, PLR platelet-to-lymphocyte ratio, IG Immature granulocytes

Laboratory parameters of the patients enrolled at admission or pre-operation

SG Strangulated group, NSG No-strangulated group, M $\pm$ SD mean $\pm$ standard deviation, WBC white blood cell, NEU neutrophils, NLR neutrophil-to-lymphocyte ratio, PLR platelet-to-lymphocyte ratio, IG Immature granulocytes

Logistic regression analysis

In binary logistic regression analysis, vomiting and WBC and IG counts were independent risk factors for predicting strangulated adhesive SBO (vomiting, OR 3.92, 95%CI 1.10–14.02, $P < 0.05$; WBC, OR 2.11, 95%CI 1.18–3.76, $P < 0.05$; IG count, OR 15.41, 95%CI 8.07–31.59, $P < 0.001$). And their VIF were 1.029 (vomiting), 1.115 (WBC), and 1.145 (IG count), separately. all of them less than 5, showing no serious multicollinearity problem. Prolonged time from symptom onset to hospital significantly decreased the likelihood of strangulation (OR 0.89, 95% CI 0.82–0.96, $P < 0.05$). Abdominal distention, tenderness, and

levels of NEU and NLR were not associated with intestinal strangulation ($P > 0.05$) (Fig. 2). Fig. 2 Logistic regression analysis for predicting strangulated adhesive SBO (* $P < 0.10$)

Logistic regression analysis for predicting strangulated adhesive SBO (* $P < 0.10$)

ROC curve and AUC analysis

ROC curve analysis was performed to evaluate the laboratory parameters for predicting strangulated adhesive SBO. As shown in Fig. 3a; Table 3, the discriminative performance was ranked as follows: IG count (AUC 0.946, 95%CI 0.92–0.97) > WBC (AUC 0.784, 95%CI 0.73–0.84) > NEU (AUC 0.756, 95%CI 0.70–0.82) > NLR (AUC 0.655, 95%CI 0.58–0.73). The IG count can be considered as the optimal predictive utility with a cut-off of $0.305 \times 10^9/L$ ($P < 0.001$), yielding a sensitivity of 82.60%, specificity of 100.00%, LR+ of ∞ , and LR- of 0.17 (Table 3). In the subgroup analysis of onset time to hospital (Fig. 3b), the IG count had the best predictive value for strangulated adhesive SBO within 6 h from onset to hospital (AUC 0.961, 95%CI 0.90–1.00, Sen 89.5%, Spe 100%), and the cut-off value was $0.280 \times 10^9/L$ ($P < 0.001$). Fig. 3 ROC curves for A IG and other inflammatory biomarkers (WBC, NEU, NLR), and B IG within 6 h from onset time to hospital

Parameters	AUC (95%CI)	Cut-off	Sen (95%CI, %)	Spe (95%CI, %)	LR+	LR-	P-value
IG (all time)	0.946 (0.92, 0.97)	0.305 ($\times 10^9/L$)	82.60 (76.0, 88.20)	100.00 (95.40, 100)	∞	0.17	< 0.001
IG (<6 h)	0.961 (0.90, 1.00)	0.28 ($\times 10^9/L$)	89.50 (76.00, 95.80)	100.00 (97.10, 100)	∞	0.11	< 0.001
WBC	0.784 (0.73, 0.84)	14.05 ($\times 10^9/L$)	45.50 (36.90, 53.10)	100.00 (95.30, 100)	∞	0.55	< 0.001
NEU	0.756 (0.70, 0.82)	8.80 ($\times 10^9/L$)	64.70 (57.40, 72.00)	71.10 (61.10, 81.10)	2.24	0.50	< 0.001
NLR	0.655 (0.58, 0.73)	4.75	50.90 (43.30, 58.70)	75.90 (66.60, 85.20)	2.11	0.65	< 0.001

SBO small bowel obstruction, AUC area under the curve, 95%CI 95% confidence interval, Sen sensitivity, Spe specificity, LR+ positive likelihood ratio, LR- negative likelihood ratio, IG immature granulocyte, WBC white blood cell, NLR, = neutrophil-to-lymphocyte ratio, Inf infinity

ROC curves for A IG and other inflammatory biomarkers (WBC, NEU, NLR), and B IG within 6 h from onset time to hospital

AUC and cut-off values for strangulated adhesive SBO

SBO small bowel obstruction, AUC area under the curve, 95%CI 95% confidence interval, Sen sensitivity, Spe specificity, LR+ positive likelihood ratio, LR- negative likelihood ratio, IG immature granulocyte, WBC white blood cell, NLR, = neutrophil-to-lymphocyte ratio, Inf infinity

Study internal validation

We used K-fold cross validation ($K = 10$) to perform the internal validation of IG count for predicting strangulation. The model showed a high average accuracy of 90.8% ($SD \pm 6.8\%$) with a good performance and acceptable stability.

Discussion

Our study is the first to establish the clinical utility of peripheral blood IG count for early prediction of strangulated adhesive SBO. Our findings demonstrated that an IG count of $> 0.305 \times 10^9/L$ strongly indicated small intestinal strangulation, with sensitivity of 82.60% and specificity of 100.0%. Crucially, within 6-hours from onset to hospital, IG count maintained exceptional sensitivity (89.50%) with a cut-off value of $0.280 \times 10^9/L$. This may effectively address the predictive void for detecting early stage intestinal ischemia.

In clinical practice, SBO is mostly caused by adhesions. Although only 3–8% of cases progress to strangulation, it has a high morbidity associated with delayed management of early identification and timely surgical intervention [20]. Abdominal CT serves as the primary imaging modality for detecting strangulated adhesive SBO and can recognize advanced intestinal ischemia before operation. However, it exhibits limited sensitivity for identifying mild-to-moderate ischemia and injury at an early stage [21]. Additionally, a transition zone on imaging cannot accurately predict surgical urgency [22]. Therefore, the

temporal advantage of serum biomarker changes preceding imaging findings is particularly significant for early clinical detection of intestinal strangulation. It is well known that when intestinal ischemia occurs, the body can rapidly release pro-inflammatory cytokines (e.g., interleukin-6 (IL-6) and tumor necrosis factor- α (TNF- α)), which can stimulate bone marrow mobilization of IG into the peripheral circulation within 2 h, later, they transform into mature segmented neutrophils in the bone marrow within 7–10 days [23, 24]. It largely preceded conventional biomarker elevation (WBC, NEU, and lactate). Our study has confirmed that during the early stages of disease onset (particularly within 6 h), an IG count level exceeding $0.280 \times 10^9/L$ was highly suggestive of intestinal strangulation. The sensitivity of 89.50% and specificity of 100.00% surpassed those reported in studies on IG count predicting mesenteric ischemia (74.1% and 98.2%, respectively) [14] and intestinal volvulus necrosis (76.2% and 85.7%, respectively) [13]. In addition, the study by Friziero et al. suggested that NLR remained an independent predictive factor for ischemia in adhesive SBO, with 78% sensitivity and 65% specificity [25]. New evidence also suggested that high levels of serum procalcitonin can sensitively detect intestinal strangulation [26] and predict surgical treatment needs [27]. However, both of them remained normal within 6 h of intestinal ischemia. Thus, IG count can be considered as an early (< 6 h), and highly specific serum biomarker for identifying intestinal strangulation to improve intestinal survival rate timely. Thereby it can increase patients' survival rate, reduce complications, and improve patients' quality of life.

Equally important, IG count is also less cost of money and time and more easy to obtain than CT and other blood markers. IG count is one common parameter of complete blood count with a lower detection cost of less than \$ 2.8, which has a higher cost advantage than 128 CT scan with a cost of \$ 98.58 in China. It greatly reduces the financial burden of patients and medical insurance expenditure. Furthermore, the majority of clinical studies also have focused on building clinical prediction models of intestinal strangulation, which has poor clinical applicability. Owing to its multiple evaluation parameters, which require a lot of time to detect, collect, and analyze variables [28–30]. Undoubtedly, it takes more time to obtain than IG count.

As we seen, IG count had a high predictive value and specificity for ultra-early stage (< 6 h) of intestinal ischemia and strangulation, providing a faster bedside assessment tool than CT. Previously in 2024, our group explored the clinical predictive value of IG count for right-sided complicated acute diverticulitis (cAD) [19], while this study about strangulated adhesive SBO, may break new ground by demonstrating predictive value of IG count for surgical acute ischemic disease, thereby paving the way for its broader clinical application. In the study of infectious diseases, such as sepsis, acute appendicitis, and pancreatitis [31–34], an elevated IG count reflected the compensatory response of the bone marrow to bacterial infection and correlated positively with pathogen load. In contrast, in our study of intestinal ischemia and strangulation in adhesive SBO, elevated IG count was the result of tissue hypoxia directly activating the bone marrow granulocyte pool significantly earlier than the secondary infection stage. This findings broke through the limitations of traditional views of IG count as an infection marker. Moreover, the early identification of strangulation in adhesive SBO hold significant clinical value, especially within 6 h from onset time to hospital, more importantly, by enabling timely surgical intervention to rescue ischemic bowel and save lives, which differently from by only mitigating the risk of antibiotic misuse in cAD. Surprisingly, this study showed that sensitivity of IG count for predicting bowel strangulation at an early stage (< 6 h) was as high as 89.50%, even higher than in cAD with a sensitivity of 80.60%. Of course, the prerequisite of IG count for predicting severity of the primary disease must be based on the disease's type, and then can build a rigorous clinical hierarchical evaluation system of severity in future.

The main limitations of this study lied in its retrospective design and sample size from a single center source, introducing the potential for selection bias and information bias. Firstly, it was lack of whole data about time of onset to hospital during all hospitalizations to dynamic assessment. We only performed logistic regression analysis instead of survival analysis of the time dimension. Secondly, it lacked a formal cost-benefit analysis from being conducted due to incomplete cost information and clinical follow-up. Thirdly, it may limit the generalizability of our results. Although post hoc analysis indicates sufficient statistical power for the primary outcome measure, the power may be insufficient for certain subgroup

analyses. Fourthly, the inconsistency of IG standards may limit its clinical application due to different testing methods of IG counts among different institutions. Fifthly, due to ethical considerations and routine clinical practice, the diagnosis of NSG was based on the absence of signs of ischemia on imaging and a favorable response to conservative management, without pathological confirmation by surgery. On the one hand, it introduced the classification bias for NSG group may contain some undetected cases with transient or occult ischemia. On the other hand, it was potentially subject to verification bias, as the definitive diagnostic standard (surgical exploration) was applied predominantly to the most severe cases, which may lead to an overestimation of the diagnostic accuracy of IG counts. Sixthly, the time from symptom onset to hospital mainly relied on patients or their families to recall symptom onset times to hospital, which may have inaccuracies. Besides that, the high specificity(100.00%) of IG in our study may have potential biases, such as co-infections, which we were unable to identify. Given the above limitations, future multicenter, prospective studies are expected to validate our findings in real-world settings.

Conclusions

Peripheral IG count might have a novel clinical value for predicting strangulated adhesive SBO, with 89.5% sensitivity and 100.00% specificity, especially within 6 h from onset to hospital. In the future, it might be a prospective early additional parameter to recognize intestinal strangulation, rather than a replacement.

Abbreviations

Small bowel obstruction

Computed tomography

White blood cell count

Neutrophil

C-reactive protein

High

-sensitivity CRP

Homocysteine

Immature granulocytes

Xinxiang medical university

Granulocyte colony-stimulating factors

Non-steroidal anti-inflammatory drugs

Strangulation group

Non-strangulation group

Neutrophil-to-lymphocyte ratio

Platelet-to-lymphocyte ratio

Mean $\pm$ standard deviation

Odds ratios

95% confidence intervals

Receiver operating characteristic

Area under the ROC curve

Variance inflation factor

Sensitivity

Specificity

Positive likelihood ratio

Negative likelihood ratio

Interleukin-6

Tumor necrosis factor- α

Complex acute diverticulitis

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Acknowledgements

No.

Authors' contributions

Q. S. participated in the study design, collection, and data analysis. Y. C. participated in data collection and analysis and drafted the original manuscript. Q. W. participated in the revision of the manuscript and supervision of all processes. All authors have read the final manuscript and have approved its submission.

Funding

This study was supported by a grant from the Henan Provincial Science and Technology Research Project(no. LHGJ20210547).

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This study was reviewed and approved by the Ethics Committee of the Third Affiliated Hospital of the Xinxiang Medical University (XXMU). And the requirement of written informed consent from patients was formally waived by Ethical committee of the Third Affiliated Hospital of XXMU, for the reasons that this study performed the retrospective review of de-identified patient data and ensure consistency with institutional approvals.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

References

本 PDF 由 PMC 开放获取全文自动生成，供课题研究学习使用；排版以出版商版本为准。